

Trainings ▾

General Information

Note : There are two different piston and liner combinations available for this engine platform. The ferrous cast ductile (FCD) piston must be used in combination with a nitrided cylinder liner. The steel piston must be used in combination with a high boron liner. It is important that these parts are used in this combination stated and **never** mixed. An engine must **never** be run with a mixture of FCD and steel pistons. It must contain **only** a full set of FCD pistons or a full set of steel pistons. The liners can be distinguished from each other by appearance. The nitride liners will have a matte appearance on the outer circumference. The high boron liners are bright and shiny in appearance.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

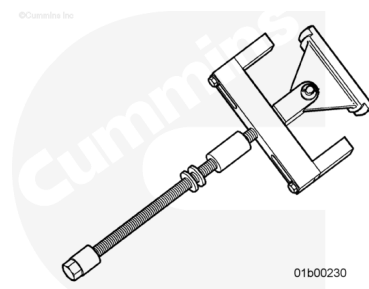
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. Refer to the equipment manufacturer service information.
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)
- Remove the piston and connecting rod. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)

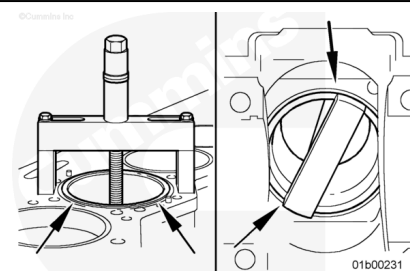
Remove

Use the cylinder liner puller, Part Number 3163745, and puller plate, Part Number 3162886, to remove the cylinder liner.

Wind down the threaded rod to extend the arm. Turn the arm to the side to enable passage through the liner.

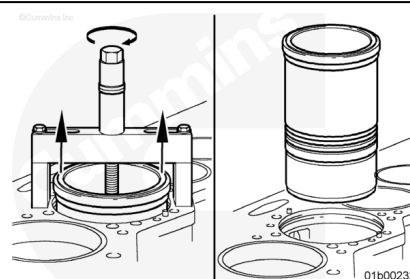


Install the cylinder liner puller, Part Number 3163745, in the cylinder liner. The puller feet **must not** touch the top of the liner. The puller arms **must** be positioned firmly on the bottom of the liner.



Turn the puller screw until the liner loosens in the block.

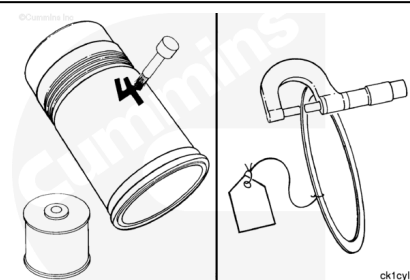
Remove the tool and the liner.



Use a liquid metal marker to mark the cylinder number and bank on each liner. Mark the cylinder liner on the camshaft side of the cylinder liner.

If sealing rings were used, use a tag to mark the cylinder number.

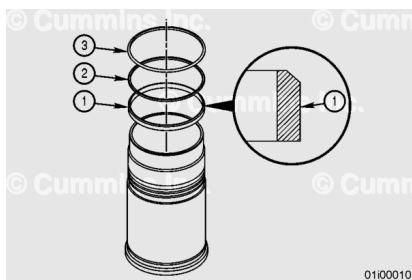
Measure in several places and record the thickness of the sealing rings used in each cylinder. The thickness of the sealing ring is one factor in determining liner protrusion. This information **must** be known when the liners are installed in the engine.



Disassemble

Remove and discard the two D-rings (2 and 3).

Remove and discard the crevice seal (1).



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a wire brush. Flying debris and dirt can cause personal injury. Make sure the wire brush is rated for the correct rpm if the brush is motor-driven.



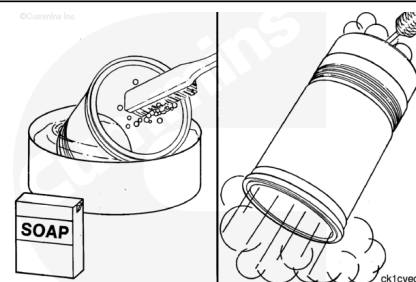
⚠ CAUTION ⚠

Do not use a hone, aluminum oxide cloth, or sandpaper to clean the cylinder liner bores. Abrasives can damage the finish and the pattern and can contaminate the liner.

Use a high-quality steel wire brush to clean the liner flange seating area and the secondary press fit below the liner flange.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

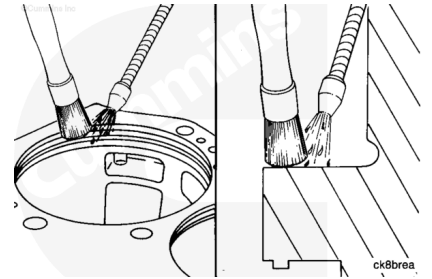
Use a nonmetallic brush, detergent soap, and warm water to clean the inner diameter of the cylinder liner.

Clean the liners with solvent or steam.

Dry the cylinder liners with compressed air.

⚠ WARNING ⚠

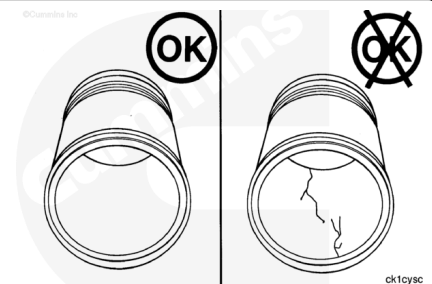
When using solvents, acids, or alkaline material for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the bottom of the cylinder liner flange with safety solvent.

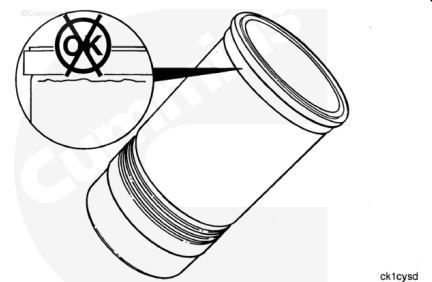
Note : Inspect the cylinder block counterbore. Refer to Procedure 001-027 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-027.html)

Inspect the liners for cracks on the inside and outside diameters.



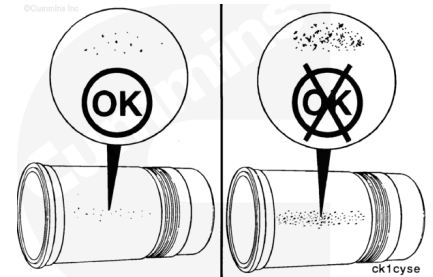
Inspect for cracks under the flange.

Cracks can also be detected by using either magnetic inspection or the dye method.



Inspect the outside of the liner for excessive corrosion or pitting. Pits **must not** be more than 1.6 mm [0.063 in] deep.

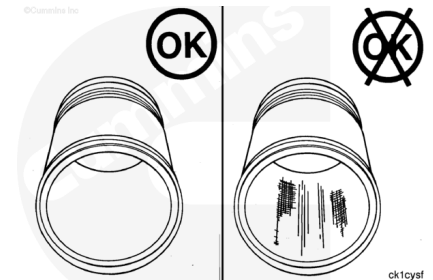
Replace the liner if the pits are too deep or if the corrosion can **not** be removed with a fine emery cloth.



Inspect the inside of the liner for vertical scratches deep enough to be felt with a fingernail.

If a fingernail catches in the scratch, the liner **must** be replaced.

Inspect the inside of the liner for scuffing or scoring.

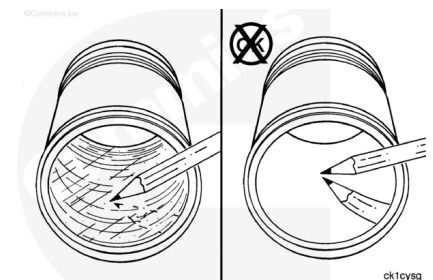


Inspect the inside of the cylinder liner for liner bore polishing.

A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of etch patterns.

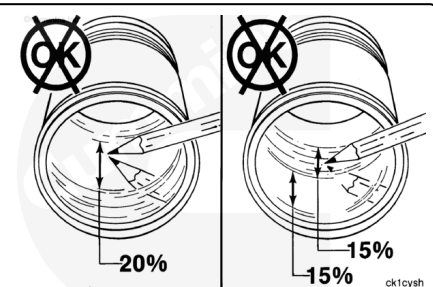
A heavy polish produces a bright mirror finish in the worn area, with no traces of hone marks or an etch pattern.

For further information on liner bore polishing, use the Parts Reuse Guidelines, Bulletin 3810303.



Replace the cylinder liner if:

- A heavy polish is present over 20 percent of the piston ring travel area.
- Thirty percent of the piston ring travel area has both moderate and heavy polish, with one-half of the 30 percent (15 percent) consisting of heavy polish.

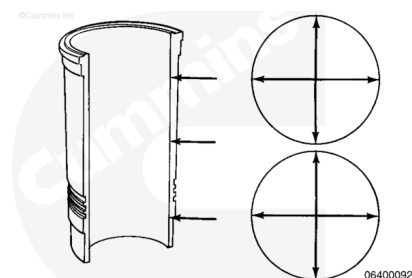


Measure

Use a dial bore gauge. Measure the inside diameter of the cylinder liner at the top, the bottom, and the middle of the piston ring travel area. Perform two measurements at each location. The measurements **must** be 90 degrees apart.

Cylinder Liner Inside Diameter

mm		in
158.75	MIN	6.250
158.78	MAX	6.251



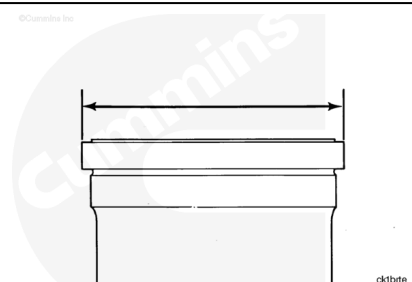
Note : The maximum liner ovality for all positions is 0.04 mm [0.0016 in].

If the cylinder liner is **not** within specifications, the cylinder liner **must** be replaced.

Measure the cylinder liner flange outside diameter.

Upper Press Fit Cylinder Liner Flange Diameter

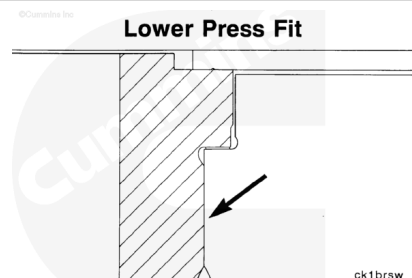
	mm		in
Standard	190.31	MIN	7.493
	190.36	MAX	7.495



If the cylinder liner is **not** within specifications, it **must** be replaced.

The liner design incorporates a press fit between the upper liner bore and the area of the liner directly below the liner flange. This is referred to as the “lower press-fit” design.

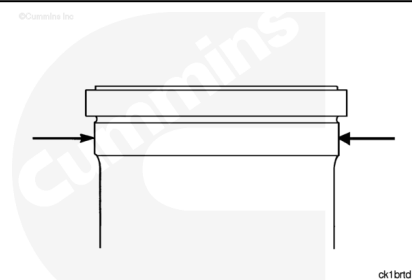
Cylinder liners with standard and oversize press-fit diameters are available.



Measure the lower press-fit outside diameter.

Lower Press-Fit Area Outside Diameter

	mm		in
Standard	181.82	MIN	7.158
	181.86	MAX	7.160



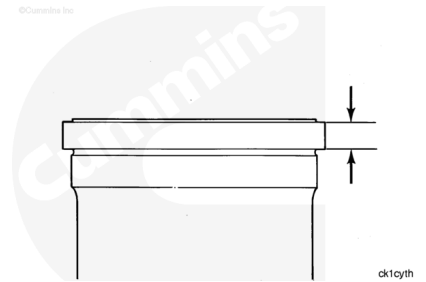
Measure the cylinder liner flange thickness.

Cylinder Liner Flange Thickness

mm		in
13.35	MIN	0.525

Cylinder Liner Flange Thickness

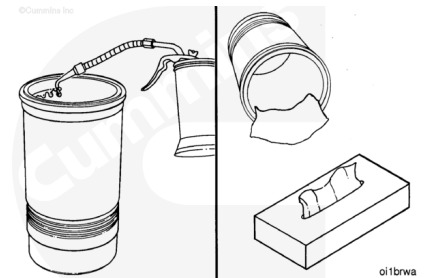
mm		in
13.40	MAX	0.527



Apply a thick film of clean engine oil to the bores of the liners for final cleaning. Leave the oil on for 5 to 10 minutes.

Use a clean, lint-free paper towel to wipe the oil from the bores until the black and gray deposits are removed.

Do **not** place the liners in an area where dirty airflow can contaminate the liners.



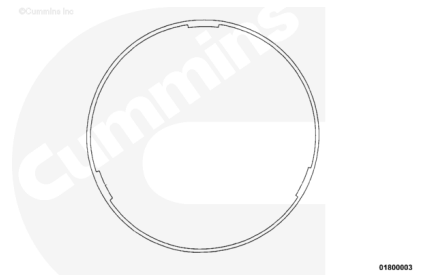
Assembly

The seal rings have three locating tabs on the inner diameter. The tabs have an interference fit to the liner lower press-fit diameter to hold the seal ring in place during liner installation.

Install the seal rings.

The seal ring **must** be straight on the liner upon installation. Use finger pressure to push the seal ring near the tabs to fit the seal ring down and over the lower press fit diameter during installation.

This practice during installation of the seal ring will prevent deformation that will result in the seal ring **not** fitting squarely on the bottom of the liner flange.



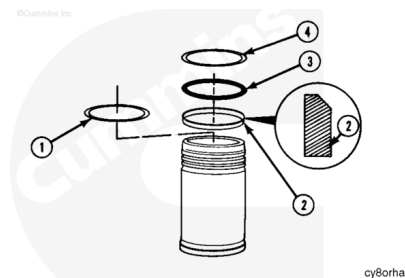
Cylinder Liner Seal Rings

Nominal Thickness	Number of Notches
0.45 mm [0.0180 in]	Two
0.52 mm [0.0205 in]	Two
0.55 mm [0.0220 in]	Two
0.66 mm [0.0240 in]	Two
0.78 mm [0.0310 in]	Two

Install the crevice seal. The beveled-edge of the crevice seal (2) **must** be positioned as shown.

Some o-rings have a "D" shape cross section. This type of o-ring **must** be installed with the flat side against the cylinder liner.

Install the o-rings in the position shown. Use the mold mark on the o-ring to check whether the o-ring is twisted.



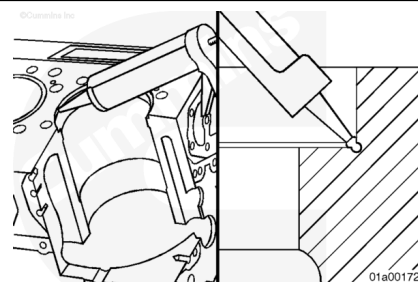
1. Liner, counterbore sealing ring
2. Crevice seal
3. Dark blue o-ring
4. Orange o-ring.

Install

Apply a 2 mm [0.079 in] bead of gasket sealant, Part Number 4918771, to the corner of the cylinder block counterbore.

Make sure that the end of the bead overlaps the beginning of the bead, approximately 6 mm [0.236 in].

Install the cylinder liner within 15 minutes of applying the sealant.

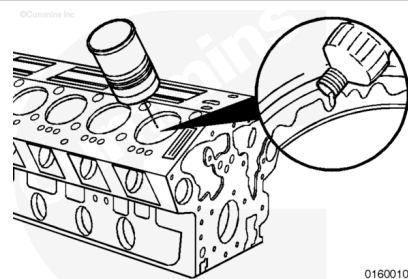


⚠ CAUTION ⚠

Do not use petroleum-based oil. This will cause swelling of the packing ring.

Use vegetable oil to lubricate the inside diameter of the packing ring bores.

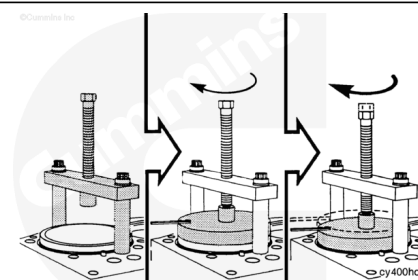
Use hand pressure to push the cylinder liners into the block.



Use liner installation tool, Part Number 3163329, or equivalent, to install the cylinder liner. Install the bridge assembly. Install two cylinder head capscrews.

Torque Value: 68 n•m [50 ft-lb]

Install the push plate. Make sure it is aligned correctly in the liner. Turn the pusher screw until it touches the plate.



Use a wrench. Do **not** apply more than 135 n.m [100 ft-lb] of torque. Turn the pusher screw until the liner flange touches the counterbore ledge.

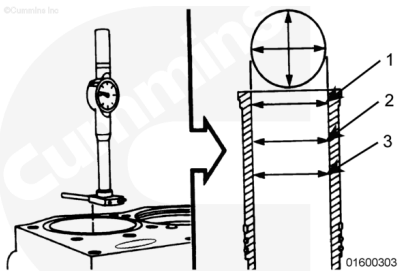
Remove the tool.

Check the liner protrusion. Use the following procedure for instructions on how to measure liner protrusion. Refer to Procedure 001-064 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-064.html)

Use a dial bore gauge. Measure the inside diameter of the cylinder liner at the top, bottom, and middle of the piston ring travel area.

Perform two measurements at each location. The measurements **must** be 90 degrees apart.



New Clinder Liner Inside Diameter			
	mm		in
Positions 1, 2, and 3	158.74	MIN	6.250
	158.78	MAX	6.251

Note : The maximum liner ovality for positions 1, 2, and 3 is 0.04 mm [0.0016 in].

If the cylinder liner is **not** within specifications the liner **must** be replaced.

Inspect for a twisted o-ring.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html>)
- Install the cylinder head and related components. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.